

Tran Quoc Truong

Applied AI Engineer · AI Agents & RAG · Backend Systems

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Summary

Applied AI Engineer building agentic applications, RAG and retrieval systems, LLM evaluation infrastructure, and Python backend platforms, with a focus on reliability, observability, and practical delivery.

Experience spans enterprise code intelligence, trace-based AI assurance, model fine-tuning and serving, quality engineering, and CI-driven delivery; independently develops Omni-Agent as a modular platform for knowledge, agents, and evaluation.

Technical Skills

AI & Agentic Systems: RAG, Hybrid Retrieval, LangGraph, MCP (Model Context Protocol), Tool Calling, Context Engineering, LiteLLM, Qdrant, pgvector, Docling

LLM Evaluation & Observability: LLM Evaluation, LLM-as-a-Judge, OpenTelemetry (OTel), Runtime Trace Reconstruction, Regression Testing, Deterministic Quality Gates, Langfuse

Backend Engineering: Python, FastAPI, Django/DRF, Pydantic, SQLAlchemy, PostgreSQL, Redis, Celery, SSE / Streaming APIs, Pytest, SQL, React, TypeScript

ML, Fine-Tuning & Model Serving: PyTorch, Hugging Face Transformers, PEFT, Unsloth, LoRA/QLoRA, vLLM, BitsAndBytes Quantization, Synthetic Data Generation

Infrastructure & Delivery: Docker, Docker Compose, Kubernetes, Google Cloud Platform, RunPod, Nginx, MinIO, GitLab CI/CD, Azure DevOps Pipelines, Linux

Experience

FPT Software , Applied AI Engineer	Ho Chi Minh City, Vietnam Dec 2024 – present
- IQP (Code Intelligence & Retrieval): Built hybrid code retrieval combining dense search, BM25, and code-graph relationships across enterprise repositories, exposing repository-aware engineering context to AI clients through Model Context Protocol (MCP). - Agent Assurance / Flezi Polaris: Designed and implemented the core trace-based Agent Assurance engine using LiteLLM and OpenTelemetry capture, multi-turn trajectory reconstruction, behavioral evaluation, and reproducible regression gates; the capability was later presented by the Flezi Polaris team and received the Golden Solution Prize (1st place) at IVS Solution Day 2.0 2026. - Automation & Delivery Engineering: Developed automated validation across API, database, web, and mobile layers and integrated repeatable execution into GitLab and Azure DevOps CI pipelines. - Banking Systems Verification: Built reusable cross-platform automation and SQL-based validation across banking and payment systems, reducing manual regression effort by approximately 70%; an omnichannel automation proof of concept contributed to approximately 20 person-months of follow-on delivery work.	
FPT Software Academy , Full-Stack Developer Intern	Ho Chi Minh City, Vietnam Sept 2024 – Dec 2024
- Built the backend for an online learning and assessment platform using Django with role-based access control and automated exam workflows. - Integrated a FastAPI computer-vision service for AI-assisted real-time exam proctoring and face detection. - Containerized the multi-service architecture with Docker Compose, Nginx, and Cloudflare Tunnel.	

Achievements

Golden Solution Prize (1st Place) — IVS Solution Day 2.0 2026 Flezi Polaris team solution built around the Agent Assurance capability for trace-based AI-agent evaluation and governed quality verification.	Sept 2026
2nd Place — IVS Hackathon 2026 Designed and implemented the core Agent Assurance engine independently; awarded 2nd place for trace-based behavioral evaluation and governed regression gates.	Aug 2026

Second Prize — Autonomous Driving Research Paper Competition

Apr 2024

Co-authored YOLOv8, lane-segmentation, and PID-control research; completed the simulation benchmark in 125.8 seconds with a 100% score.

Featured Projects

Omni-Agent — Applied AI Platform (Knowledge Builder & Agent Assurance)

A modular Applied AI platform combining versioned document knowledge pipelines, agent runtimes, and agent evaluation within a project-centric multi-service architecture.

- Platform Architecture: Designed a modular multi-service architecture with project-scoped control plane, isolated domain services, asynchronous workers, PostgreSQL, Redis, S3-compatible object storage, and containerized deployment.
- Knowledge Builder: Built a versioned document pipeline covering source ingestion, Docling parsing/OCR, chunking, Qdrant indexing, citation-grounded retrieval, and MCP-based knowledge access.
- Agent Runtime & Integration: Implemented LangGraph-based agent workflows with tool calling, context-aware retrieval, MCP integration, and SSE streaming for multi-turn execution over project knowledge.
- Agent Assurance: Integrated trace-based evidence capture, evaluation, regression, and quality-gating capabilities into the platform.

APIT — Agent Programmatic Integration Testing

AI-assisted API testing framework generating structured test scenarios from API documentation via fine-tuned LLMs.

- Developed a bilingual synthetic dataset generation pipeline producing 1,258 domain-specific API testing samples.
- Fine-tuned Qwen2.5-3B with Unsloth and PEFT LoRA/QLoRA, achieving macro-F1 of 0.655 (+41.5% relative improvement over Llama-3.2-3B baseline).
- Deployed the fine-tuned model on RunPod through an OpenAI-compatible vLLM service with BitsAndBytes quantization and runtime LoRA adapters.
- Integrated pgvector-backed cosine-similarity retrieval and a hybrid evaluation pipeline combining LLM-as-a-Judge, fuzzy matching, and deterministic JSON-schema validation.

E-Commerce AI Assistant — Conversational RAG & Search

End-to-end shopping assistant combining filtered vector search and multi-agent conversational retrieval.

- Built product semantic search using Milvus, cosine similarity, embeddings, and dynamic category/price filters.
- Implemented multi-turn conversational agent with LangGraph and LangChain for context-aware product consultation.
- Containerized multi-service architecture with FastAPI, PostgreSQL, Redis, Milvus, and Docker Compose.

Education

FPT School of Business & Technology , Master of Software Engineering in AI	Ho Chi Minh City, Vietnam May 2026 – present
FPT University , Bachelor of Artificial Intelligence	Ho Chi Minh City, Vietnam 2021 – 2025

Languages

- Vietnamese:** Native
- English:** Professional working proficiency